

Maths

Unit 2 - Numbers
and Sequences

WALT:

6Nc.04 – Use knowledge of square numbers to generate terms in a sequence given its position.

6Ni.08 – Use knowledge of multiplication and square numbers to recognise cube numbers.

WALT: Use knowledge of square numbers to generate terms in a sequence given it's position. Use knowledge of multiplication and square numbers to recognise cube numbers.



**With support, I
can answer
questions about
square and cube
numbers.**

**Independently, I
can answer
questions about
square and cube
numbers.**

**I can complete the
“Think Like A
Mathematician”
investigation.**

WALT: Use knowledge of square numbers to generate terms in a sequence given it's position. Use knowledge of multiplication and square numbers to recognise cube numbers.

Retrieval Practice

Here are five digit-cards.

1

6

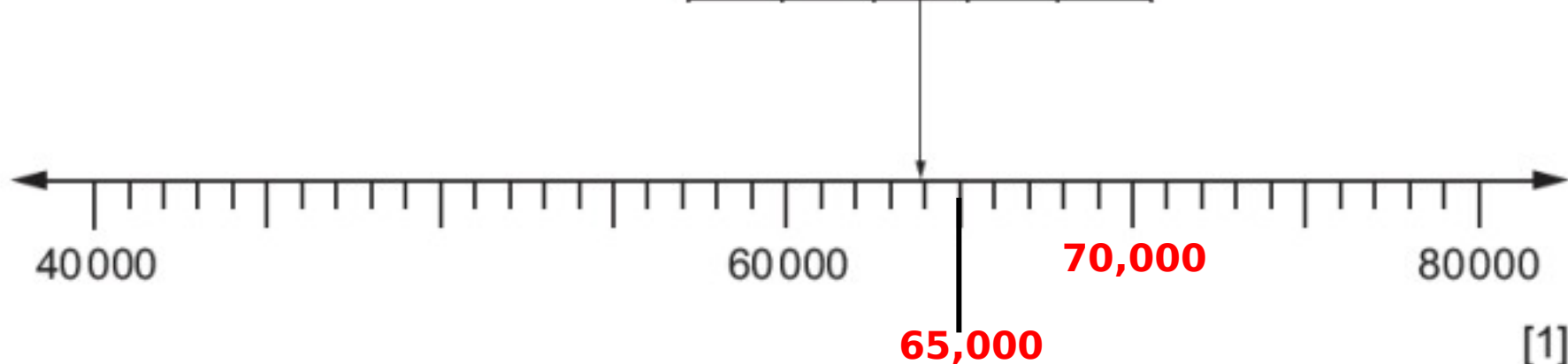
3

5

9

Use **three** of these cards to estimate the missing number on the number line.

6 3 9 0 0

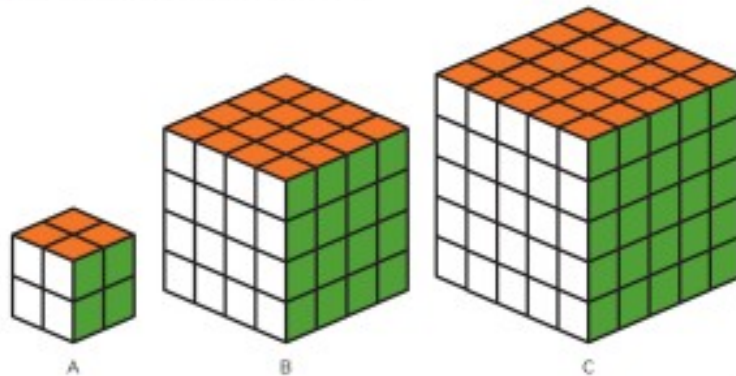


WALT: Use knowledge of square numbers to generate terms in a sequence given it's position. Use knowledge of multiplication and square numbers to recognise cube numbers.

Pg. 28 - 29

Exercise 2.2

- What is the ninth square number?
- Calculate.
a 5^2 b 10^2 c 7^2
- Find two square numbers that total 45 when added together.
- Here are three cubes of increasing size.



How many small cubes are in each of the large cubes?

Finding the number of small cubes in the large cube is equivalent to finding the volume of the large cube. Both answers are the cube of the length. What is equivalent to finding the square of the length?

- Calculate.
a 5^2 b 1^2 c 3^2
Check your answers to questions 4 and 5 with your partner.

2.2 Special numbers

- Copy this Carroll diagram and write a number less than 100 in each section.

	Odd	Not odd
Cube number		
Not a cube number		

- Find two cube numbers that total 152 when added together.
- Classify these expressions into two groups. Explain how you chose the groups.

2^3 3^2 $2^3 + 1$ half of 4^2 $3^2 - 1$

Discuss your answer with your partner.

Think like a mathematician

Two consecutive squares

1 and 4 are two consecutive square numbers. $1 + 4 = 5$

4 and 9 are two consecutive square numbers. $4 + 9 = 13$

Investigate the sums of two consecutive square numbers. If you are systematic you should find an interesting pattern.

Adding odd numbers

The first two odd number are 1 and 3. Their sum is 4

The first five odd numbers are 1, 3, 5, 7 and 9. What is their sum?

Investigate the sums of consecutive odd numbers starting at 1. What do you notice?

WALT: Use knowledge of square numbers to generate terms in a sequence given it's position. Use knowledge of multiplication and square numbers to recognise cube numbers.

1 What is the ninth square number? **81**

2 Calculate.

a 5^2 **25**

b 10^2 **100**

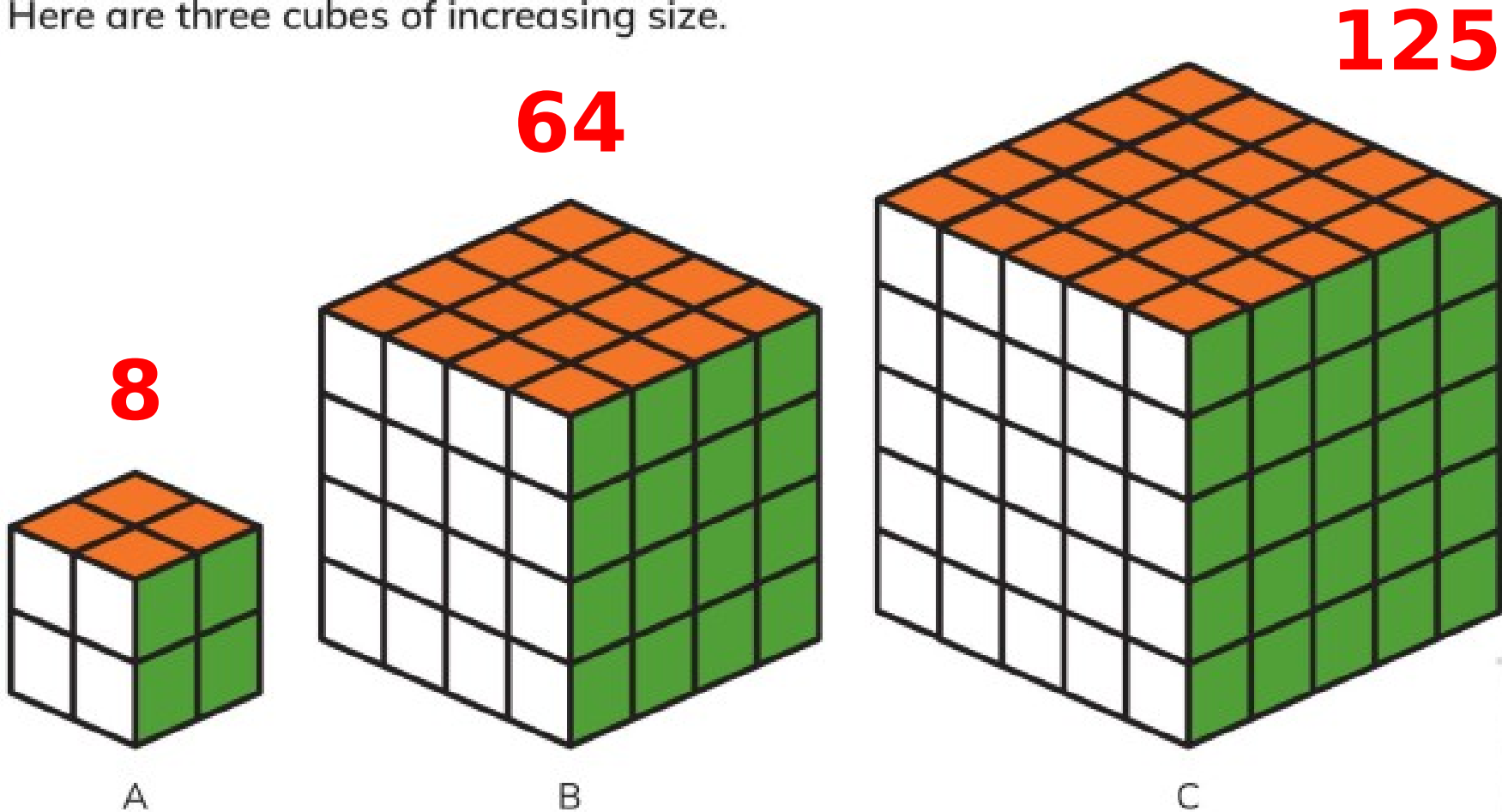
c 7^2 **49**

3 Find two square numbers that total 45 when added together.

$36 + 9$

WALT: Use knowledge of square numbers to generate terms in a sequence given it's position. Use knowledge of multiplication and square numbers to recognise cube numbers.

4 Here are three cubes of increasing size.



How many small cubes are in each of the large cubes?

WALT: Use knowledge of square numbers to generate terms in a sequence given it's position. Use knowledge of multiplication and square numbers to recognise cube numbers.

5 Calculate.

a 5^3 **125**

b 1^3 **1**

c 3^3 **27**

Check your answers to questions 4 and 5 with your partner.

WALT: Use knowledge of square numbers to generate terms in a sequence given it's position. Use knowledge of multiplication and square numbers to recognise cube numbers.

- 6 Copy this Carroll diagram and write a number less than 100 in each section.

	Odd	Not odd
Cube number	1 or 27	8 or 64
Not a cube number	3, 5, 7 ... 25, 29 ...	2, 4, 6, 10 ... 62, 66 ...

- 7 Find two cube numbers that total 152 when added together. **125 + 27**



- 8 Classify these expressions into two groups. Explain how you chose the groups.

2^3 3^2 $2^3 + 1$ half of 4^2 $3^2 - 1$

Discuss your answer with your partner.

WALT: Use knowledge of square numbers to generate terms in a sequence given it's position. Use knowledge of multiplication and square numbers to recognise cube numbers.



Think like a mathematician

Two consecutive squares

1 and 4 are two consecutive

4 and 9 are two consecutive

Investigate the sums of two

If you are systematic you will

Adding odd numbers

The first two odd numbers

The first five odd numbers

Investigate the sums of consecutive

What do you notice?

Adding odd numbers

$$1 + 3 = 4$$

$$1 + 3 + 5 = 9$$

$$1 + 3 + 5 + 7 = 16$$

$$1 + 3 + 5 + 7 + 9 = 25$$

The sum of the first two odd numbers $= 2^2$

The sum of the first three odd numbers $= 3^2$

The sum of the first four odd numbers $= 4^2$

The sum of the first five odd numbers $= 5^2$